



Department of Electrical Engineering

Graduation Project II

Giving Voice to Silence: Smart Glove for Sign Language Translation

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STUDENT DECLARATION OF OWN WORK

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Abstract

In everyday social interactions, communication obstacles are a major barrier for deaf and mute people, especially when interacting with individuals who do not understand sign language. To address this challenge, this project proposes the design of a smart wearable glove that converts hand gestures used in sign language directly into audible speech, eliminating the need for mobile applications or external devices.

During the development of the proposed system, several design approaches were considered. Vision-based solutions using cameras and image processing techniques were explored; however, they were found to be sensitive to lighting conditions, computationally complex, and unsuitable for real-time portable applications. Smartphone-based applications were also examined, but these solutions reduce system independence and require continuous user interaction. Consequently, a sensor-based wearable system was selected due to its portability, low power consumption, and ability to provide real-time responses.

The proposed solution consists of a smart glove equipped with flex sensors to detect finger bending and an inertial measurement unit (IMU) to capture hand orientation and motion. An ESP32 microcontroller is used as the central processing unit because of its compact size, low energy consumption, and sufficient processing capability for wearable systems. Predefined hand gestures are mapped to corresponding audio outputs using an audio playback module, which plays the translated speech through a speaker. This approach enables direct conversion of sign language gestures into speech without relying on intermediary text displays.

Although the primary focus of this project is system design and analysis, the proposed architecture demonstrates strong feasibility, scalability, and suitability for real-world applications. The system is expected to deliver fast response times, minimal power consumption, and enhanced user independence. In conclusion, the proposed smart glove represents an effective assistive technology solution that can significantly improve communication for deaf and mute individuals, with substantial potential for future implementation and development.

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1.0 Problem Statement

People with special needs face difficulties in social life, especially since we live in communities that do not attempt to learn sign language. This creates a barrier to communication with others in society and leads to loneliness and isolation. Therefore, there is a need for a smart system that helps translate hand gestures into audible speech instantly, in order to facilitate communication and enable individuals with special needs to express their needs.

2.0 Project Objectives

The project aims to design a smart hand that helps hearing- and speech-impaired individuals communicate easily with others in society through sensing technologies and intelligent processing. The objectives of this project include:

1. Analyzing hand gestures and converting them into digital data.
2. Translating the gestures into speech.
3. Improving communication between hearing- and speech-impaired individuals and people who do not understand sign language.
4. Designing a practical, user-friendly, comfortable, and low-cost system.

3.0 Literature Survey

Sign language is an important way of communication for deaf and speech-impaired people. It depends mainly on hand gestures, finger bending, and hand orientation to express letters and meanings. Each gesture has a specific shape that must be recognized correctly. Smart glove systems work by converting these hand movements into electrical signals using sensors. Flex sensors are used to measure how much each finger bends, while motion sensors such as accelerometers or IMUs are used to detect hand movement and direction. A microcontroller processes this data and translates the gestures into text or speech.

In [1] introduced a smart glove that translates American Sign Language (ASL) letters into text and speech. Their system uses five flex sensors and an accelerometer connected to an Arduino Nano. The recognized gestures are sent to a mobile phone using Bluetooth, where the translated text and sound are displayed. The system showed a response time of about 0.6 seconds, which proves that wearable gloves can be used for real-time sign language translation.

However, there are several shortcomings of this technique. Firstly, this process only converts alphabets, but not full sentences or words; moreover, some letters like U and V have similar finger configurations and thus are hard to distinguish. Moreover, using wires makes the device uncomfortable, while the incorporation of the mobile phone and Bluetooth reduces its independence and increases the energy requirement of the system.

Both the existing technique and the proposed smart glove share common characteristics, including flex sensors and hand motions for detecting letters. Contrary to some previous techniques that solely concentrated on alphabet recognition, the current project focuses on the detection of predefined words.

The difference between the proposed system and the reference material is in the microcontroller. The present system incorporates ESP32 microcontroller which is faster and consumes less power than Arduino Nano. In addition, the present system uses

Existing smart glove technologies show that sensor-based wearable systems are effective for sign language translation. Compared to camera-based systems, smart gloves are easier to use, less

affected by lighting conditions, and more suitable for daily use. These technologies support the development of a simple, low-power, and user-friendly smart glove.

The following literature is established on reference [2], which describes the use of a sensor-based method for converting sign language to text and speech through a smart glove and an Android mobile application.

The most widely recognized form of communication regarded in the deaf and hearing-impaired community is sign language. Sign language utilizes hand gestures and finger movement to spell out words and phrases. As most people do not know sign language, it becomes a challenge for most deaf people to be able to communicate freely.

To overcome this communication barrier, sign language translation technology has been developed that translates sign language into text and speech. Recent innovations in wearable technology and embedded systems have enabled sign language interpretation by using intelligent gloves with sensors that can detect and identify human gestures. Sensor-based systems are preferable due to their portability and less dependence on environmental conditions.

The existing work done in sign language translation systems has almost followed a pattern of either vision-based systems or sensor-based systems. In vision-based systems, images are taken by cameras, and image processing is done to recognize the signs. The systems are able to yield good results; however, they require proper lighting and fixed camera positions.

Sensor-based systems utilize wearable technology such as smart gloves equipped with sensors such as flex sensors and accelerometers to measure finger bending and hand movement. As described in reference [2], sensor information can be processed by a microcontroller and transmitted to a smartphone application where translated words can be displayed and vocalized. Although these benefits exist, the current systems also have a few shortcomings. In most of the current systems, the translated text is displayed or the speech is produced using a smartphone or a computer, making the systems less independent. Moreover, the systems are limited in that they support a limited number of gestures since the recognition is based on predetermined mappings. The expansion of the systems tends to increase the costs.

In comparison to the translation systems that utilize smartphone technology, the proposed project targets a more independent and portable device that can be worn. By incorporating the output audio into the glove itself, the system becomes more portable and convenient for translation. As can be seen from reference [2], sensor-based smart gloves form a successful solution for sign language interpretation. Nonetheless, the disadvantages represented by previous interpretations show that there remains a requirement for a relatively independent and friendlier solution, thereby justifying the rationale for forming a smart glove project.

4.0 Projects Realistic Constraints

1. Performance

device must translate hand movement with high accuracy and no delay because response time for sensors and speed processing are very important to achieve excellent and reliable device.

2. Economic Constraint

device must be affordable cost so people with hearing and speech impairment can buy it. It's required to balance between choosing components, performance and cost.

3. Functionality

the hand gesture must be user-friendly and convenient by covering set of hand signs that used a lot by people to have convenient communication.

Additional features such as wireless and natural voice output must be integrated because it provides the user with more natural and acceptable experience.

4. Energy Constraints

The smart glove relies on a tiny battery, so saving power really matters. It's packed with sensors, plus things like a speaker or a screen, and all of them draw energy. To keep the glove running smoothly, we need to use low-power parts and write efficient code. That way, the glove does its job without draining the battery too fast, and you're not stuck charging it all the time.

5. Environmental Constraints

The smart glove should be able to handle environmental factors like changes in temperature, humidity, and dust. It should be lightweight, and able to resist any condition the patient or the user may face. All components must be secured so that it can be comfortable with hand movement and bending. The glove also must work safely, without overheating or causing pollution. Any electronic waste from the project should be managed responsibly.

6. Health and Safety

Because the device is a wearable device and it is directly on the user's hand, health and safety considerations are very important. To reduce the possibility of electric shock, irritation, or discomfort, all electronic components must be adequately insulated. To avoid hand strain during extended use, the glove should be flexible and lightweight. Electronic heat generation must remain within safe bounds, and skin-friendly materials must be used. In addition, clear safety guidelines during connecting, especially handling batteries, wiring, and soldering activities if it needs, must be followed.

5.0 Alternative Designs

Based on the reviewed article [1], an alternative design for a sign language translation system is proposed, which relies on a wearable smart glove. This design aims to facilitate communication between deaf or speech-impaired individuals and people who do not understand sign language by converting hand movements into textual or audible output.

The proposed system employs **flex sensors** attached to the fingers of the glove to measure the bending of each finger during sign language gestures. Flex sensors were selected due to their simplicity, lightweight design, and ease of integration with the glove, making the system comfortable for daily use over extended periods.(see Figure 1 for the design layout and display screen showing recognized letters).

Data from the sensors are transmitted to a processing unit, where **machine learning algorithms** analyze the input. The system utilizes a pre-trained machine learning model based on a large dataset of sign language gestures. This model allows the system to accurately recognize different hand movement patterns and associate them with specific gestures.

Once a gesture is recognized, the system converts it into a readable text or audible speech output. This approach significantly improves communication, particularly for individuals who do not know sign language. Supporting voice output further enhances the naturalness and usability of the interaction.

Experimental results reported in the article indicate that the system achieves a high recognition accuracy, approximately **94%**. However, challenges remain, such as expanding the number of supported gestures and improving the system's ability to distinguish between similar gestures. In summary, the alternative design demonstrates that a smart glove based on flex sensors combined with machine learning is an effective and straightforward solution for sign language translation. This design also provides a solid foundation for future enhancements, including the integration of additional sensors and the optimization of processing algorithms to increase accuracy and efficiency.



Figure 1: Layout of the smart glove design with a display showing recognized letters [1].

6.0 The Design Solution

The suggested design solution of the smart glove will include a wearable sensor system that will convert the sign language gestures into speech. It should be noted that the main goal of this system is providing assistance to deaf-mute people who have difficulty communicating with those people who are unfamiliar with sign language. This design choice emphasizes the following features of the system: portability, low power consumption, quick response, and user-friendly performance [3].

The system comprises multiple hardware and software units that form an embedded system together. Five flex sensors have been incorporated into the fingers of the glove to measure the bending of the fingers during the gesture actions [4]. The flex sensors have been chosen due to their light weight and flexibility. Moreover, an accelerometer and gyroscope sensor (MPU6050)

have also been used in the embedded system to recognize the orientation and movements of the hand for improved gesture recognition.

As the primary microcontroller chosen for the project, the ESP32 microcontroller was chosen due to several advantages provided by this chip, including its compactness, low energy requirements, fast processing, built-in Wi-Fi and Bluetooth features, and low costs. The use of the ESP32 microcontroller is preferable when compared with larger processing chips like Raspberry Pi due to their better suitability for wearables [5].

Regarding the output mechanism, the design uses an audio player module that includes the DFPlayer Mini audio player coupled with a speaker. The DFPlayer Mini stores pre-recorded sound information in microSD cards. Once the system recognizes a particular gesture, it automatically outputs the voice message from the DFPlayer Mini. This approach was chosen over the smartphone output for improved portability.

While implementing and integrating the system's hardware, a few technical difficulties emerged. The first difficulty arose in relation to the MPU6050 library setting on the Arduino IDE. At the start of the development process, no required MPU6050 libraries were available, causing compilation errors in the program coding. It was then decided to use the Adafruit MPU6050 library that was more compatible with the ESP32 microcontroller.

Another technical problem occurred when testing the flex sensors. Indeed, the sensors constantly gave zero outputs, especially for thumb and index finger flex sensors. However, after investigating the problem within the circuit, it became clear that such behavior was caused by the lack of voltage division via fixed resistors. Furthermore, it had to be improved, since some ESP32 analogue inputs generated electrical noise. Such difficulties were overcome by changing the wiring design and minimizing jumper wires.

Further complications arose from the need for hardware calibration and stability of the sensor values. Flex sensors had slightly varied analogue sensor values even when bent identically, making it imperative to adjust the threshold values for each flex sensor individually. Moreover, there was occasional noise in the readings of the MPU6050 due to vibrations from the shaking

hands as well as instability of the sensor placement inside the glove. This issue was overcome using software filtering and retesting.

The resulting system integrates the sensor, signal processing, and output components into an easily portable unit that can translate pre-specified gestures made through sign language into speech. The proposed approach makes possible a compact system that is relatively inexpensive to develop and implement yet still maintains good performance.

6.1 Design Integration

The design integration of the smart glove system focuses on combining hardware components with embedded software to create an efficient and reliable wearable device. The primary objective is to ensure seamless cooperation between all subsystems, including sensing, processing, communication, and output, so that sign language gestures can be translated into understandable speech or text in real time, thereby improving communication with deaf and hard-of-hearing individuals [1] [6].

To achieve this objective, the glove incorporates carefully selected components based on performance, practicality, and suitability for wearable applications. The sensing subsystem relies on five **flex sensors** positioned along each finger to measure finger bending during hand movements [1] [7]. These sensors were selected due to their durability, flexibility, and reliability in wearable environments. Their analogue output allows direct interfacing with the microcontroller, eliminating the need for complex signal-conditioning circuitry and simplifying system integration [8].

From the software point of view, the system uses an inbuilt software application that analyzes the data generated by the sensors at all times. Inbuilt software analysis allows for interpretation of specific hand movements and translating them to specific outputs. The software analyzes the data generated by sensors and compares it with threshold values to determine the specific hand movements made.

In addition to finger movement detection, the glove integrates an **MPU6050** accelerometer and gyroscope to capture hand orientation and motion data [9]. This sensor provides precise six-axis motion tracking while maintaining a compact size and low power consumption, making it well suited for lightweight and comfortable wearable devices.

The ESP32 microcontroller serves as the core processing unit of the system, handling data acquisition, processing, gesture recognition, and output control [10]. The decision to use the **ESP32** instead of a more powerful platform such as the Raspberry Pi was made deliberately. The ESP32 offers several advantages for wearable embedded systems, including significantly lower power consumption, which extends battery life, integrated Wi-Fi and Bluetooth connectivity for optional wireless communication, and fast real-time processing suitable for low-latency gesture recognition. Furthermore, its compact size, minimal heat generation, lower cost, and ease of integration make it safer, more comfortable, and more reliable for wearable use.

In contrast, although the Raspberry Pi provides higher computational power, it is less suitable for this application due to its larger physical size, higher power consumption, increased heat generation, and reliance on an operating system. These characteristics limit its practicality in a wearable glove, making the ESP32 a more balanced choice in terms of efficiency, responsiveness, and physical suitability.

For output generation, the system includes a **DFPlayer Mini** audio module connected to a small speaker to play prerecorded voice messages corresponding to recognized gestures [8]. This module was selected for its low power requirements, ability to store audio files locally on a microSD card, and simple serial communication with the ESP32.

The embedded software integrates sensor calibration, signal filtering, gesture recognition algorithms, and output coordination into a unified workflow. Gesture data is received by the ESP32, compared against predefined gesture patterns, and then translated into the appropriate audio or textual output.

Overall, the design integration ensures that every component supports the primary goal of the smart glove: to provide a compact, efficient, and user-friendly wearable system capable of translating sign language into clear speech or legible text in real time.

6.1.1 Components



Figure 2 : Esp Microcontroller

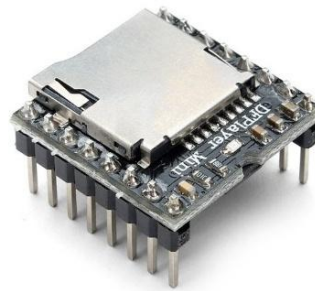


Figure 3:DF player mini

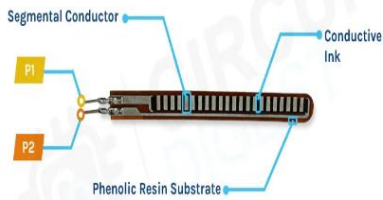


Figure 4: Flex sensor



Figure 5: MPU6050

6.2 Design Verification

The proposed design solution includes the development of a smart glove that can recognize predefined sign language movements and turn them into speech. The key goal behind the project is to help the deaf-mute people communicate with people who do not know sign language. Portability, low energy consumption, high-speed response, simplicity, cost-effectiveness, and sufficient accuracy are some of the characteristics to be taken into consideration during the design phase [3].

The hardware and software modules have been combined into one compact device. Five flexible sensors will be mounted on the glove fingers to track finger bending during sign language performance. The selection of these sensors was made due to their light weight, flexibility, low price, and applicability in the context of wearable electronics. Finger positions detected by sensors give important data for further recognition of different gestures.

In order to increase accuracy of recognition, the design includes a sensor called MPU6050 that integrates three axes of accelerometer and gyroscope together. With this sensor, it becomes possible not only to sense finger bending but also orientation and movements of hands. Such characteristic plays a crucial role because some gestures have the same values of flex sensor data but different hand positions and motions [4].

A decision to use ESP32 microcontroller as a processor of the system was taken for the following reasons: ESP32 is small, consumes less energy, operates quickly, has wireless communication capabilities, and its price is relatively low. As compared to bigger processing systems like Raspberry Pi, ESP32 appears to be a better choice for wearable devices since it provides enough computational power and energy efficiency [5].

The gesture recognition algorithm used in the project is based on sensor fusion along with threshold-based classification algorithms, rather than complex machine learning algorithms. This is done intentionally to simplify computation and to make sure that the process is fast enough to perform in real time using the ESP32 microcontroller platform [3]. In contrast to the other

implementation described in Section 5, in the current implementation, the smart glove uses threshold-based classification for reasons of simplicity and less complex computation processes and is thus suitable for performing in real time with ESP32.

To produce voice output, the project uses DFPlayer Mini module connected to a speaker. Recorded voice commands are saved on microSD card and then played whenever a certain gesture is identified by the gloves. Using smartphone as a way of producing voice commands was considered but abandoned because this would make the glove dependent on smartphone availability and connectivity. This way, users will have the ability to communicate using the glove independently of other devices.

Modular design principles were applied through the entire process of creation of the project. Individual components such as flex sensor module, motion sensing module, gesture recognition code, and audio output were developed and tested separately before implementing them into the total system. Such an approach facilitated troubleshooting, minimized complexity, and increased the reliability of the project's components.

A number of challenges arose during the implementation and integration process. One of the first issues faced in the process of the implementation pertained to the configuration of the MPU6050 libraries in Arduino IDE. The application of certain libraries led to errors because of the fact that they were not entirely compatible with the ESP32 platform. The challenge was overcome through the analysis of alternative libraries, and eventually, the Adafruit MPU6050 library was chosen for further use.

One more difficulty appeared when testing the flex sensors. The problem was that some of the sensors gave constant output equal to zero, especially those connected to the thumb and index finger. Investigation of the problem showed that analog voltage divider circuits needed to be connected properly. Once fixed resistors were added to make the proper configuration of the analog measurement circuit, proper output values could be read from the sensors.

Some difficulties appeared while developing the algorithm for gesture recognition. Although sensor values were detected properly depending on the finger movement, performing the same gesture more than once would give slightly different values for the sensors because of differences in hands sizes, positions of fingers, bending strength, gloves fit, user movement, etc. Moreover, there were some gestures where the ranges of values for sensors overlapped, and therefore they could not be classified properly. This problem was solved through numerous calibration experiments performed on the sensors [12].

Other problems arose in regard to stability and elimination of the noise. Small variations were experienced in the MPU6050 readings because of the vibrations created by hands and the movement of the sensor within the glove. There were also problems of electrical noise from certain analog sensors. This problem was dealt with by placing sensors appropriately, optimizing the wiring system, software filtering, and calibration.

In conclusion, various adjustments were made to the hardware design to make the system more reliable and comfortable for use. Adjustments were done to place the sensors appropriately in order to enhance consistency in reading and wiring to minimize the problem of interference.

The developed system effectively combines sensing, signal processing, gesture recognition, and audio output functions in a wearable unit. The suggested concept provides an appropriate and economical solution to translate sign language in real time without compromising portability, energy efficiency, and usability. The prototype smart glove proves that a sensor-based embedded technology can ensure effective communication support without the drawbacks inherent to visual-based technologies [3].

6.2.1 Hand Gestures



Figure 4 : I have a question



Figure 5 : Stop



Figure 6 : Ok



Figure 7 : No



Figure 8 : I am hungry



6.2.2 Sensors Readings

MPU6050 sensor readings

The MPU6050 sensor was used to detect the orientation of the hand during each gesture. Broad acceleration ranges were selected to make the system more flexible and less sensitive to small hand movements.

The flex sensors were used to measure finger bending values for gesture recognition. Each gesture was assigned a specific range based on experimental readings collected during testing.

MPU6050 Readings

Table 1 : MPU6050 Readings

Gesture	MPU6050 Range
OK	X = -6.0 to 2.0, Y = -11.0 to -7.0, Z = -4.0 to 4.0
I have a question	X = -11.0 to -7.0, Y = -5.0 to 5.0, Z = -4.0 to 4.0
Stop	X = -11.0 to -7.0, Y = -5.0 to 5.0, Z = -4.0 to 4.0
No	X = -2.0 to 5.0, Y = 8.0 to 11.5, Z = -2.0 to 5.0
I am hungry	X = -11.0 to -8.0, Y = -5.0 to 2.0, Z = -5.0 to 2.0

Flex sensor readings

The ranges were selected experimentally to improve gesture detection accuracy while reducing false detections.

Table 2: Flex Sensor Readings

Gesture\Fingers	Thumb	Index	Middle	Ring	Pinky
Ok	170–610	340–425	270–410	330–390	1570–1660
I have a question	80–180	480–800	390–560	220–400	1530–1800
Stop	600–880	1000–1300	700–1000	800–1100	1900–2200
No	380–680	280–390	300–420	200–380	1550–1730
I am hungry	70–420	390–580	290–500	270–480	1600–1820

7.0 Project Budget

The projected budget needed to execute the suggested smart glove system is shown in this section. The price of the primary electronic parts, sensors, and additional supplies are all included in the budget. The prices indicated are estimates that could vary based on market rates and availability.

Table 3: Project Cost Estimation

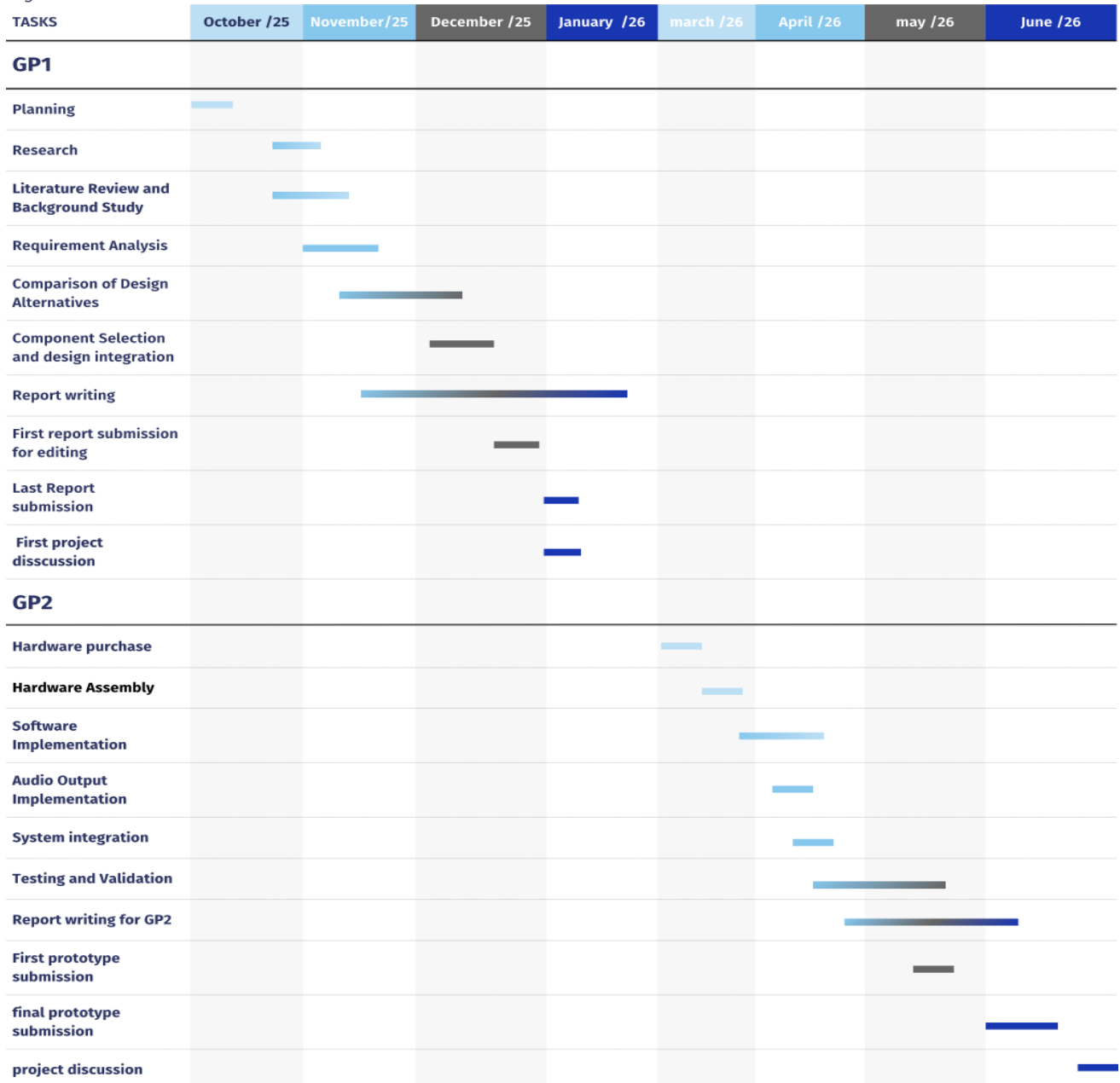
Component	Price
Esp32	7.50 JD
DF player mini	1.95 JD
5 Flex sensors	82.5 JD
MPU6050	3.70 JD
Breadboard	0.95 JD
Speaker	0.70 JD
MicroSD card	6.00 JD
Wearable glove	1.00 JD
Jumper wires	3.00 JD
Glue	1.00 JD

Total budget = 108.30 JD

8.0 Project Schedule

The Gantt chart below illustrates the planned timeline for the graduation project, divided into two main phases: GP1 and GP2. GP1 focuses on planning, research, literature review, system analysis, design integration, and report writing. GP2 represents the proposed practical implementation phase, including hardware acquisition, system development, integration, testing, and final documentation. The chart demonstrates the logical sequence of tasks and their expected duration throughout the project lifecycle.

Figure 9:Gantt chart



9.0 Conclusion

The current project involved the designing, implementing, and evaluating a smart glove that was to aid deaf and speech impaired persons to communicate with those who cannot interpret sign language. This device consists of five flex sensors and an MPU6050 sensor, which are used to detect the movements of fingers and the positioning of hands respectively. The signals detected are interpreted by the ESP32 microcontroller, which translates certain defined gestures to voice commands using the DFPlayer Mini audio module and speakers.

Various issues arose during the entire development process, which included calibration of sensors, noise interference, problems related to MPU6050 library, and accuracy of gestures' detection. Such issues have been sorted out through various modifications on the part of hardware as well as software.

Finally, the last system proved that a glove with sensors is capable of being a cheap, portable, and power efficient device for translating sign language in real time. The employment of a threshold gesture recognition scheme provides fast execution and high reliability without increasing computational load.

Generally, the project successfully fulfilled its major goals by providing a device for translating predefined gestures into voice. Potential developments might be aimed at expanding the range of supported gestures, increasing recognition efficiency by implementing more complex recognition algorithms, adding support for wireless communications, and decreasing the size of the device.

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